

Playful Co-Design: Creating an AR-Prototype with Nurses in Interlocking Remote and On-Site Workshops

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ABSTRACT

Deeply engaging nurses in a participatory co-design process, especially in times of COVID-19, is challenging. In this case study, we shed light on the process of developing a prototype for AR-glasses in nursing. We show the challenges we faced, the methods we used and how they contribute to the core principles of participatory design. A special focus is laid on small-scale interventions with high-impact, that helped us to truly engage users. We introduce empathetic ways to connect contrasting work environments, establish mutual understanding, make the abstract more graspable with playful tools like PLAYMOBIL®, and support co-design development with online formats. Finally, we discuss the transferability to other projects.

CCS CONCEPTS

• **Human-centered computing** → **Participatory design**; *Field studies*; • **Applied computing** → *Health care information systems*; • **Computing methodologies** → *Mixed / augmented reality*.

KEYWORDS

Participatory Design, Co-Design, Design Thinking, Hospital Care, Augmented Reality

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1 INTRODUCTION

Even before the COVID-19 pandemic, nurses in German hospital care were confronted with high workload and information overload due to rising patient numbers [8] and increasing legal and economic demands [33, 40]. Staff shortages worsen the situation and lead to emotional exhaustion [40]. Augmented reality (AR) glasses are a promising technology for facilitating care processes [31] and are

increasingly addressed in applied scientific research with practical partners [18, 29, 39]. However, in order to develop AR-applications creating real value for users, nurses need to be involved in the development process.

Following this idea, the Participatory Design (PD) approach goes beyond user-centered design. Since the 1970s' workplace democracy movement in Scandinavia, PD stands for ensuring that users of information technologies play a critical role in their design [32]. According to [13], PD comprises the following guiding principles: 1) *equalising power relations* so that less recognized voices are heard, 2) *situation based actions* that are understandable rather than abstract, 3) *mutual learning* by encouraging and enhancing the understanding of different perspectives, 4) *tools and techniques* that help express needs and visions, 5) *alternative visions* about technology, and 6) *democratic practice* among those who represent others.

Putting these core concepts into practice is challenging [3]. The community argues that PD might have lost some of its clarity and identity as it is often reduced to a banal form of user-centered design without really empowering users to participate in decision making and co-creation [3]. In addition, many scientific studies lack an adequate description of the methods used. Systematic literature reviews reveal that only a few authors explicitly used arguments directly related to PD guiding principals to reason how participants were recruited and why methods and tools were used for what outcome in electronic healthcare interventions [27, 38]. Furthermore, out of 23 publications on AR in nursing lately identified by [39], only three describe methods identifying use cases. That shows, that Bannon's call "to be more daring when it comes to proposing and scrutinizing technological alternatives, from the point of view of their usefulness to people, not merely as new business cases" is hardly answered [3, p. 32].

One reason could be that the healthcare sector presents a number of challenges in terms of participatory introduction of modern technologies [1, 34, 38]. Strong hierarchy gaps between clinic professions make it difficult to balance power relations and to give nurses enough space to make their voices heard. Yet, shift work and limited time resources complicate the organization for common appointments which emphasizes the need of representatives within democratic practice. Bearing in mind that the nursing profession is not a desk job, working with digital tools and having unlimited access to a PC cannot be taken for granted.

In the context of new and hard-to-imagine technologies, such as AR, it is even more challenging to enable participation on eye-level: The possibilities and limitations of these new technologies must be communicated clearly and comprehensibly so that future users

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can discuss their use competently. Ideation and prototyping need to be implemented with simple and ideally well-known tools that enable user participation and co-creation. A systematic literature review shows that while most studies point out the importance of low fidelity prototyping, there are few methods that have been used and studied for AR use cases that are suitable for technically unequipped and untrained users [10]. Storyboards, paper-prototypes or Wizard-of-Oz scenarios were used to illustrate and test ideas and concepts [10].

These methods are well established in face-to-face workshops, introducing a further challenge: the COVID-19 pandemic. This still has a strong impact on the availability of participating nurses. Joint on-site appointments can only be conducted in small groups and under strict hygienic conditions. High sick leave and quarantine meet a dramatically increased workload and push many participants to their limits.

Considering these challenges, the question arises whether and, if so, how participation is possible in this difficult context. In this case study, we took a detailed look at the participatory process of introducing AR-glasses into the everyday care of two German hospitals. In the following, challenges, methods used and contributions to PD principles are described. A special focus is laid on small-scale interventions with high-impact, that helped to truly engage users while meeting institutional dependencies and demands.

2 CASE SET-UP

The research project PARCURA (funding 2019 to 2023) addresses the participatory introduction of data glasses in hospitals to improve the quality of work for nursing staff. The reported activities were strongly affected by the COVID-19 pandemic, as they took place 2021 to 2022.

Interdisciplinary stakeholders included: representatives of nursing professionals, technical experts (User Experience (UX), AR and hospital infrastructure) and organizational specialists (nursing science and hospital administration). As the primary users, nurses were the focus of this case study.

Two diverse German hospitals were equally involved: One hospital was located in a more rural area with fewer staff, less specialized wards, and a comparatively low-equipped IT-infrastructure that forced nurses to continue working with paper documents. The other hospital was located in the centre of a larger city and was digitally well equipped. In the selection of participating nurses, emphasis was placed on heterogeneity by covering different experience levels, age groups and genders. The extend of active participation varied and depended on COVID-19 regulations, availability and phase of the design process.

Bringing together a larger round of care professionals required a great deal of organizational effort and appropriate lead time. Planning reliability for participation was difficult to achieve due to high sick leaves. Consequently, appointments had to be scheduled long in advance while considering short-term cancellations. Furthermore, external guests were partially prohibited from entering the hospital due to COVID-19 regulations. Contact bans also allowed only a limited group size in workshops. Impressions of the workshops are shown in figure 1.

The IT-infrastructure in both hospitals was highly restricted. Access to the internet was limited by firewalls and external devices were not allowed within the intranet. Nurses on the wards did not have unlimited access to a dedicated PC and did not have their own professional e-mail addresses or business phone numbers.

3 METHOD AND PARTICIPATORY STRATEGY

According to the PD-guidelines and the project limitations, a suitable structure had to be found to actively involve the nursing staff. In this section, we first provide insights into team composition, followed by a closer look at the co-designers as a central role in the project. Afterwards, we introduce a recurring pattern and the framework we have been guided by as well as the methods and tools that were used.

3.1 Teams and Co-Designers

To work flexibly and efficiently in a democratic practice, we established a funneled communication pattern in which representatives played a crucial role as they became increasingly involved. All employees of both hospitals were regularly informed about the progress of the project and asked to provide feedback (1st level participation). Five nurses per hospital formed a team that actively participated in smaller and less regular activities, like workshops, and represented their hospital (2nd level participation). This group was accompanied by a nurse who built the bridge between the hospital and other stakeholders as a co-designer, who participated actively and continuously (3rd level participation). A similar structure was used to create funnel communication within the other fields of stakeholders (e.g. the UX design team). One additional person actively coordinated the communication between all stakeholders.

A well-established method in participatory practice for actively involving users in the design process is co-design [30]. Instead of being a passive object of study, the user is considered an expert, who plays a decisive role in knowledge development, ideation and concept creation [35]. Co-designers not only express their own opinion but also represent the attitudes of their group. Following the design thinking method “Extremes and Mainstreams” [16] both co-designers differed in terms of their experience. One co-designer was new to the hospital context, coming from a nursing background in a retirement home, while the other had many years of work experience in the current hospital. These two perspectives were important to cover the needs of novices as well as long-established employees.

As mentioned earlier, time, space and technical equipment might hinder nurses from active participation. To enable both, working in the project and still on the ward with other nurses, these positions were partially funded by the project and advertised for application. In their project time, these nurses were equipped with an office workplace including PCs, telephone and e-mail access so that they were equally able to interact with other stakeholders. Through this, we created the basis for equalized power relations and democratic practice.

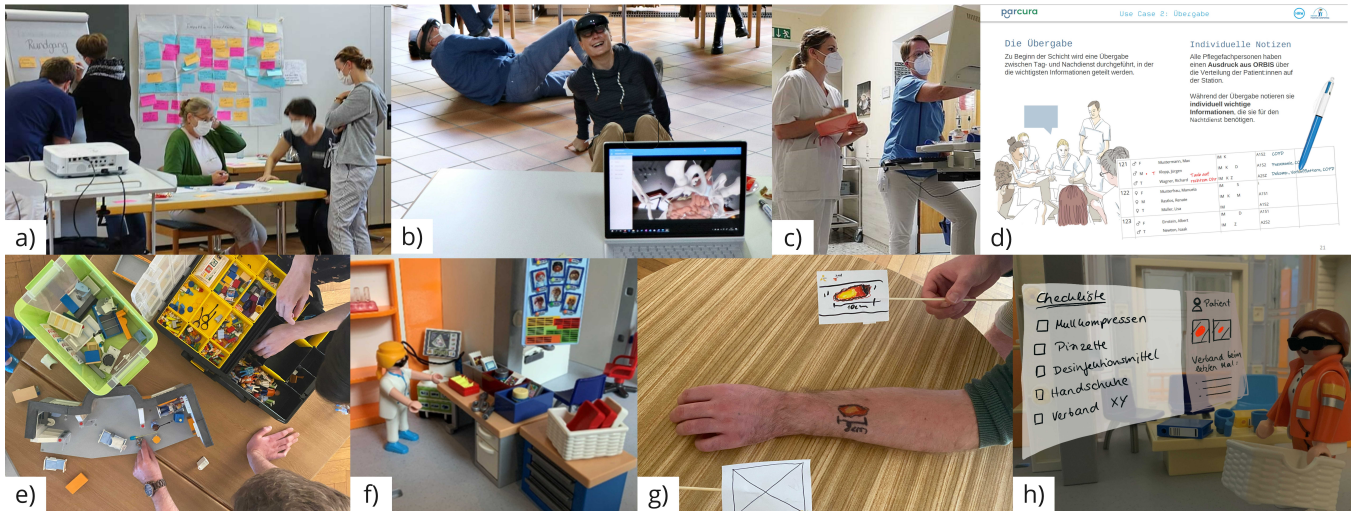


Figure 1: Impressions from different co-design workshops: a) discussion of pains and gains in nursing routines, b) exploration of AR-glasses, c) contextual inquiry in the hospital, d) extract from the persona brochure, e-f) cooperative building of scenarios in a PLAYMOBIL® hospital, g) paper-prototyping with target group, h) digital low-fidelity prototyping.

3.2 Framework and Tools

Overall, we followed a pattern called “*Diamond of Participatory Decision Making*” in discussions [19]. This moderation technique divides decision-making processes into three phases: 1) divergent thinking to gather many individual thoughts 2) discussion to contrast arguments, and 3) convergent thinking to bring results back together and make decisions. In workshops, this helped to distinguish between creative and critical thinking.

This pattern can also be found in the “*Double Diamond*” of the *Design Thinking Framework* [2, 4] used in this project. This structured and sequential approach provides orientation for the entire team and helps to draw an argumentative thread from user requirements to realization. It supports an open-ended approach and still remains goal-oriented by focusing on user needs.

Design Thinking itself makes use of a rich methodological toolbox to promote collaboration and support joint progress [2, 14, 15]. It broadens traditional UX-methods, which focus particularly strongly on the target group and their needs, like empathy maps [9], personas [5] and user journeys [7]. In order to make workshops as efficient as possible, moderation methods from the *Liberating Structures* (LS) [24] were used. These micro-structures unleash participation in groups of any size by making it possible to actively involve every participant. Their combination allows facilitators to flexibly influence group dynamics depending on the situation. Additionally, we used serious play methods within co-design as it supports creativity [17]. Based on the idea of LEGO® Serious Play® [21], we introduced the more life-like PLAYMOBIL® figurines as a potential toolkit to make the abstract more graspable in the context of AR use cases.

4 INSIGHTS AND FINDINGS

With this pattern, framework and methodological toolbox, we subdivided our journey in three phases: Understand, Explore and Develop.

For each phase, we provide insights about the main challenges, the interventions and tools, the effects, and the lessons learned.

4.1 Understand

The main aim in this phase was to establish communication on eye level and reach mutual understanding. It was crucial for the team to establish a common ground on domain specific knowledge (AR-technology and nursing profession) and to get into cooperation. To achieve these goals, we faced the following challenges:

4.1.1 Online Meetings under Unequal Conditions. The first challenge was to establish suitable communication channels for all participants. In order to meet pandemic regulations, the first attempt was to establish online meetings with a video conferencing tool. Yet, unlike the co-designers and stakeholders, the other nurses only had the opportunity to participate in such an event from home with their private devices. Compared to the other participants, nurses were not familiar with videoconferencing and therefore were in an unequal position. As a result, nurses were reluctant to engage in discussion and feedback. Accordingly, we changed the communication channel: Within the hospital teams, we aimed for face-to-face communication in small groups, as far as possible within contact restrictions.

4.1.2 Knowledge Transfer in Alternating Workshops. The second challenge was to coordinate the workshops in presence. As it was not possible to get both of the hospital teams at one table due to pandemic restrictions, we conducted workshops with alternating participation. This issue was reported by [28] as well, who came up with a similar approach of organizing workshop series in which the generated insights were rotated among the hospitals. In each case, previous results formed the basis for discussion and further processing. This strategy was followed in several workshop sessions: initial attitudes and requirements towards the use of AR in nursing were collected, different models of AR-glasses were introduced to

the hospital staff and typical use cases in nursing were mapped out in workshops. As the moderator of the workshop series stayed consistent, knowledge could be passed on reliably.

4.1.3 Overcoming Hierarchy Gaps with the Conversation Café. Third, we wanted to collect diverse and detailed information on attitudes towards AR in nursing. Due to the poor experience from the first online meeting, we laid special emphasis on equalizing power relations and ensuring that every voice was heard. Therefore, we used the LS “Conversation Café” [23] to overcome a strong hierarchy gradient in one of the on-site workshops (N=8, participants with no nursing background, nurses, team lead, professor of nursing sciences). By using this method, everyone was given the exact same time box after another to first state their own opinion and then to reflect on what they just heard. Afterwards, they were invited to discuss openly before they were again instructed to communicate their take-aways one after the other. In comparison to the virtual meeting, the willingness to speak was much better, participants opened up during the workshop and revealed worries and fears. The prompt to reflect other opinions seemed to increase mutual understanding and to take into account alternative perspectives leading to a more detailed overall vision. Hence, we assessed this method as a small-scale intervention with high impact.

4.1.4 Sharing AR-Experiences via Screen Cast. A key takeaway from the previous workshop was that nurses complained that they had a “strange feeling” about using AR as it was too abstract and not graspable enough. They requested more trial time to better understand the technology, make themselves more comfortable with it and thus make better suggestions for future scenarios. We used applications from the App Store that allowed users to experience general interaction patterns such as gesture or gaze control (e.g. MRTK example hub [22]) and those that demonstrated use in a medical context (e.g. HoloAnatomy DEMO [37]). Due to the limitations of hospital internet access, we recommend using applications that work either without accessing the internet or to provide a stand-alone solution, e.g. with a mobile Wifi router.

Usually, only the one who is wearing the AR-glasses can see the digital content. We worked with the Microsoft HoloLens 2 and established a screen cast on a video projector to share the wearer’s experience directly with a broader audience. However, deviations in transparency, color reproduction and latency occur, as the real environment is perceived only as a video and not through the glasses. Despite these limitations, it still made the seldom in-person meetings with nurses more efficient.

4.1.5 Persona and Use Case Brochure to Scale up Mutual Understanding. In other workshops we deployed mapping techniques, like empathy maps [9] and user journeys [7], to figure out current use cases in nursing that might benefit from AR-technology. However, the depth of information gathered in this way was not far-reaching enough. Participating nurses stated that we “need to be on the ward” ourselves. Hence, we conducted a work shadowing during the night shift in both hospitals to better understand nursing processes as well as the whole work environment. Yet, it is way too time consuming for all stakeholders to engage in this way. Therefore, we compiled all findings that were new to stakeholders

without a nursing background into a detailed persona [5] and associated storyboards. In a comprehensive brochure (including 55 pages), this persona was used not only to illustrate representative hospital working conditions but also to make nine potential use cases easily understandable. It was accompanied with a personal field report [11] and spread within the hospitals, among stakeholders of the project and in university courses to set a common ground and understanding of the problem space to inspire new ideas about future AR-usage. Hence, this method was crucial for scaling up as it facilitated on-boarding and could be easily referred to for further discussions. As a positive side effect, nurses reported to feel understood and heard by all this attention. Additionally, it stimulated the discussion between both hospitals as similarities and differences became transparent.

4.2 Explore

The aim in this phase was to decide on the most promising future use case scenario in which AR-glasses could make a real impact in supporting nurses. Based on the current use cases, initial ideas should be developed and discussed.

The biggest challenge was to find an appropriate prototyping tool: (1) All participants should have equal opportunities to express their own ideas quickly and without outside help. (2) The tool should provide a visual or haptic anchor for discussion. (3) The idea must be easily communicable and transportable across the hospitals. (4) The prototype must not look finished to provide enough space for further discussion and revision.

Usually, paper-prototyping combined with a role play or Wizard-of-Oz scenario is well suited for this [10]. Yet, the complexity of the nursing environment was hard to recreate in a workshop scenario as it would have been inappropriate to hold a workshop on the ward, particularly within pandemic restrictions. Inspired by the LEGO® Serious Play® method [21], we examined options to represent the hospital in a small scale. LEGO® itself was less suitable considering the time consuming building process and the ambiguity of interpreting the rather abstract bricks. PLAYMOBIL® figurines, a widely used toy in Germany that competes with LEGO®, are more suitable as they are much more detailed.

Again, we planned interlocking workshops with both hospitals. First, we run ideation and prototyping workshops with interdisciplinary teams (2 x N=4, co-designer, nurse from the ward, AR-designer, nursing scientist). Based on the persona use cases, participants were first asked to collect initial ideas on sticky notes, then discuss these ideas, and finally select the most promising one for further iteration. Afterwards, participants were introduced to a complete PLAYMOBIL® hospital set and further equipped with pen, paper, craft materials and a tablet for photo-documentation to elaborate the idea into a descriptive concept. Observations were recorded using photographs and written notes, which were discussed with participants.

In total, three concepts resulted and were discussed with a broader, interdisciplinary audience in both hospitals (2 x N=9, co-designers, nurses, nursing scientists, technicians and representatives of hospital organization). Therefore, the PLAYMOBIL® scenes and paper-prototypes were accompanied with posters and presented in a speed dating format by a representative of the group. More details for the

development of a single use case are described in [12]. Each concept was given the same time-box for presentation and were discussed in regard of added value, possible obstacles and ideas for further development. This feedback was the basis for prioritizing one use case to be implemented as a demonstrator on the HoloLens 2.

4.2.1 PLAYMOBIL® as a Graspable Material. The PLAYMOBIL® set instantly aroused the interest of participants (“Oh look at all this stuff. Even a stethoscope.”) and quickly put them into action (“Let’s set up the nurse’s station first.”). It stimulated creativity and discussion. For example, one participant found scissors and asked if they were needed. A nurse responded that they will use it as a medication divider. Another one spotted a radio and revealed how important it is for staying awake in calm, lonely night shifts. This subtle details were helpful to immerse in the nursing situation. The haptic representation was considered as “beneficial” and the material as “totally realistic with all these small utensils”.

4.2.2 Re-Enacting Situations in Small-Scale-Scenarios. PLAYMOBIL® scenes allowed to present unusual scenarios and situations that cannot be forced and observed within a contextual inquiry or job shadowing (“You get details much better that way. The situation becomes more complete. It is much easier than just interviewing. You can ask, ‘What do you need? Where does this take place?’ and then get the situation built up.”). By using diverse PLAYMOBIL® figurines, possible effects of the AR-glasses on outsiders were highlighted. One group built a scene in which the video stream of AR-glasses was used to demonstrate a procedure to a group of trainees, rather than gathering them around a patient bed. The scale of the PLAYMOBIL® hospital was commented as well suited to having the factor “room and space” in mind while coming up with possible AR-solutions (“Even in the ward rooms there is only little space available”).

4.2.3 Evolving Playful Atmosphere. A play-atmosphere evolved, which broke down hierarchies and connected strangers and diverse groups (“I found it especially great how well and enriching the collaboration in the small group with several generations and completely different professional backgrounds worked.”). On the one hand, the experience was reported as “a nice reminder on childhood” and led to a relaxed atmosphere (“Building up slows down. You have much more time for reflection and think more about details.”). On the other hand, we observed participants being rather active, hurrying to their wards in order to collect more artefacts that might be useful for prototyping (e.g. document sheets, pill boxes) or creative, as one participant began to draw a wound on his arm to play the patient.

4.2.4 Limits and Field of Application. When it came to concrete UI-ideas for the AR-glasses, the PLAYMOBIL® scale was far too small. Participants quickly switched to classic pen and paper-prototyping combined with role play. Sticky notes were hold up in front of objects or body parts to demonstrate what information should be revealed in the AR-glasses. Stressed with the UI scaling problem, first the tablet was used to take pictures of the PLAYMOBIL® scene and then to draw the UI-idea with a digital pen onto it. After the session, participants stated that “it is all about the mix” of the tools. It allowed everyone to express their design individually. They

assumed, e.g., that role playing might be something for a “rampant” whereas PLAYMOBIL® might be more accessible for shy people.

4.3 Develop

The aim of the last phase was to develop a running prototype on the AR-glasses that can be tested in a simulated nursing scenario. It was important to closely involve the co-designers in order to not only achieve a technically well-implemented result, but also to deliver a content concept that is able to demonstrate the added value for practice realistically.

4.3.1 Interlocking Content and Design Remotely. Based on the good experiences in 4.1.2, we decided to conduct interlocking series of workshops and formed two teams. The so called “content team” (N=6, co-designers, nursing scientist, hospital organization representative, human factors researcher, UX-designer) worked together continuously and iteratively on the content concept of the prototype. In the “development team”, (N=6, UX, UI and AR-development) the results were picked up and translated into an UI. Feedback and ideas from the broader nursing community were collected in an online meeting. Two persons were in both groups to ensure information exchange.

The prototype was developed in several iterations that can be broken down to following procedure: (1) The content group created a detailed user journey map, demonstrating the process of the selected use case and their ideas for AR-usage in each step. Realistic dummy data were created by referring to a fictive but plausible patient’s case. Additionally, the co-designers collected artefacts, like reference images and screenshots taken from their current document system, to stay close to the nurse’s mental model. This procedure reminds of the on-site workshop in which snippets of forms were cut out to be used in the paper-prototype. (2) The development team picked up the results in a design sprint [20, 36] and generated ideas supported by the liberating structure 1-2-4-all [24] and the creative technique SCAMPER [6]. Six independent UI-designs evolved, differences and similarities were discussed and consolidated in a group discussion. (3) These results were shared with the nursing specialists of the content group (N=4) for feedback and to agree on the most promising approach. In an iterative process, the selected UI-approach was further elaborated in cooperation until the prototype covered all main aspects of the user journey. (4) The final UI-concept was then implemented on the HoloLens 2 by the development team using the Microsoft MRTK toolkit in Unity [25]. In parallel, a scenario for a usability test was created with the co-designers, who played a fictive patient. Figure 2 illustrates this development.

Due to the joint elaboration, both sides developed more insights and understanding for each other’s work. One co-designer stated in a newsletter article: “We were able to learn and benefit from each other. [...] I found it exciting to take a look into a world that was unfamiliar to me and to experience which processes and questions are important in programming so that the prototype becomes user-friendly.”

4.3.2 Enabling Cooperation with Digital Whiteboards. As the low-fidelity prototype continued to evolve and gained expressiveness, it was inevitable to switch to a digital format. Hence, collaborating



Figure 2: Evolution of the prototype: a) First UI variations, b) conceptual prototype and c) executable prototype on HoloLens 2

in online workshops was obvious and time-saving. Yet, to engage everyone on equal footing, the gap in technical expertise had to be narrowed first. Besides video conferencing, we introduced Miro [26] as a digital whiteboard tool for online workshops and conducted training sessions so that the co-designers and representatives of less technical fields were able to participate actively. This tool allowed both groups to express and exchange their ideas about content and UI on a common ground. From iteration to iteration, active participation increased, which was reflected by active use of sticky notes and adding content to the board. In discussions the visual presentation diminished ambiguity and suggestions for improvement could be easily demonstrated.

4.3.3 Obtaining Feedback Remotely with the Chat Avalanche. During the development process, we conducted online meetings with a broader group of nursing professionals from both hospitals (N=9) to check for correctness and to receive further inspiration. As the Conversation Café was perceived as a useful technique to encourage active participation in the face-to-face setting, it was adapted for the online meeting to give everyone the same space for comments. In a “chat avalanche” each participant was prompted to answer three questions: How often does the presented use case occur in practice, what is the perceived difficulty and what potential do they see in AR-glasses support? The first question was posted in the chat by the moderator. Participants were prompted to type their answer in the input field but to not send it. Only on command, all participants sent their answers simultaneously leading to a chat avalanche. After the first two questions, everyone was invited to scroll through the answers, reflect on their content and share their impression in a group discussion. Then the next question was posted and so on. Like in the face-to-face meeting, differences and similarities were revealed, which encouraged the discussion. In comparison, the online format was more time efficient but statements were less extensive.

5 DISCUSSION

In the above findings, we have identified ways in which true participation can be deployed in accordance with PD-principles, even in a difficult context, such as involving nurses during a pandemic or to make the abstract AR future more understandable and graspable for participants.

In the first phase (4.1) it was crucial to create a common ground where power relations were balanced. In order to involve everyone equally, it is important to consider general job conditions, like space, time, availability and technical equipment. With regard to the

nurses, participation would not have worked so efficiently if there had not been a funded position that allowed both, working at the desk and on the ward. The online participation in the small round with the co-designers worked very well. For the broad workforce, however, it was still difficult to enable optimal conditions for online participation. The willingness to create online events remained low, indicating that much more lead time and technical support needs to be factored in.

Since this way of cooperation was new to most participants, we assume that it helped to make positive experiences in a on-site workshop first (like the Conversation Café in 4.1.3) before adapting its structure in a virtual form (Chat Avalanche in 4.3.3). In both situations we experienced this method as a small-scale intervention with an high impact on participation. Yet, these observed effects on different participants should be examined in future research more deeply.

Taking a look at mutual learning, we experienced how important it was to share situation based actions that are understandable rather than abstract to connect both worlds, the technical and the nursing profession. Encounters at eye level and taking time to experience the other’s world were important to immerse, understand and develop empathy for each other. Creating a complex persona story based on job shadowing and sharing AR-experiences via screen cast contribute to this. In retrospect, nurses and designers often referred to experiences and artefacts of this phase. E.g., the persona was a common anchor for discussion in a lot of workshops. The effort was considerable, but produced results that made scaling much easier and we perceived that this had a positive influence on mutual respect and team work.

In the second phase (4.2) one methodological approach stood out in particular. The mix of PLAYMOBIL®, paper-prototyping and role playing was used effectively and efficiently to develop and discuss initial ideas in order to make a reflective decision on which path to take. All PD-principles could be covered by this method: The playful approach equalized power relations and offered opportunities even for weaker voices to be heard. Situation based actions could be re-enacted in a workshop setting and fostered understanding of different perspectives, both procedural and emotional. The tools helped to express needs and alternative visions about technology. In democratic practice, representatives of all disciplines were involved in ideation and decision making. We highly recommend to further investigate that method in future research, e.g. its potential to be immersed into a use case, to experience flow or how play effects teamwork. We assume that it can also make a valuable contribution

in other application areas such as artificial intelligence, which is similarly abstract to AR, or in projects where opposing working worlds meet and situations cannot be planned.

The turning point in the third phase (4.3) was the usage of a digital whiteboard to interlock nursing professionals and designers remotely. The digital visualization and its progress was transparent to members of both groups at any time and diminished ambiguities. With this tool, feedback was quickly exchanged and the communication between co-designers and designers became increasingly efficient leading to a fast development of the final prototype. We observed that patterns from on-site experiences (paper-prototyping) could be successfully transferred to the digital, again. In retrospect, Miro could have been applied earlier. E.g. in PLAYMOBIL® workshops, the re-enacted scenes could be photographed on-site and uploaded and edited on a Miro board, shared with a projector.

With regard to the whole participatory process, the role of the facilitator should be emphasized. As [21] already recommends for LEGO® Serious Play®, “it takes someone to guide the progression, to keep the play serious and risk free, and to ensure progression”. This can be extended to the entire design thinking process, in which useful methods, such as LS, can be deployed flexibly to involve participants as best as possible.

6 CONCLUSION

Considering all the organizational limitations, we were uncertain whether a true participatory co-design process with nurses would even be possible during the pandemic. Yet, in the spirit of the open-ended but not aimless design thinking approach, we found novel ways to engage nurses at different levels, both on-site and online, within interlocking workshops. With job-shadowing resulting in a complex persona and sharing AR-experiences within groups, we demonstrate how the previously unfamiliar working worlds meet each other to form mutual understanding, both factual and emotional. We highlighted methods such as the Conversation Café and its online adaption Chat Avalanche to equalise power relations and to make each voice heard and reflected. In this environment the playful deployment of PLAYMOBIL® closed the last gaps in understanding, inspired novel ideas and facilitated discussion. Finally, we developed a high-fidelity prototype by closely linking content generation and design sprints in online co-design, supported by a digital whiteboard. With this work we unlock new approaches for future research of small playful methods with great leverage effects. This case study contributes to the call for more descriptive justification of selected methods, sharing lessons learned in practice, and explaining how both practitioners and researchers can profitably use these methods.

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